

Skills Summary

Diagnosing Transformation-Order and Inverse-Domain Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Transformations Inside the Function

Factor the inside before you read anything off it: $f(bx - c) = f\left(b\left(x - \frac{c}{b}\right)\right)$.

- Inside (input) operations act in the *reverse* of the written order and in the *opposite* direction: stretch or compress first, then shift.
- Outside (output) operations act in the order written and in the direction written: $f(x)+2$ really is up 2.
- Horizontal factor b means a compression by $1/b$; the shift is c/b , never c .

Example: Describe the graph of $y = f(3x - 12)$.

Step 1. A shift cannot be read from an unfactored inside, so pull the coefficient of x out first.

Step 2. $3(x - 4)$ is the inside, so the graph compresses horizontally by $\frac{1}{3}$ and then shifts right 4.

Try it: Factor the inside of $y = f(2x + 10)$ and read off the shift.

check: 2(x + 5)

Skill 2 — Composing in the Right Order

In $f \circ g$ the function on the right runs first: $(f \circ g)(x) = f(g(x))$.

- Compose symbolically before substituting a number — the order is visible in the formula and invisible in the arithmetic.
- $f \circ g$ and $g \circ f$ are different functions except by accident.
- An input must be in the domain of the inner function *and* its output in the domain of the outer one.

Example: $f(x) = 3x - 1$ and $g(x) = x^2$. Find $(f \circ g)(2)$.

Step 1. The inner function g takes the input, so build $f(g(x))$ symbolically before any number goes in.

Step 2. $f(g(x)) = 3x^2 - 1$, so at $x = 2$ the value is **11**.

Try it: Same f and g : find $(g \circ f)(2)$.

check: 5

Skill 3 — Inverses and Their Domains

Swap and solve gives the formula; the graph gives the domain. The domain of f^{-1} is the range of f .

- Write $y = f(x)$, swap x and y , solve for y , then state the domain separately.
- Squaring both sides is reversible only on the side where the radical was non-negative — that is what selects one branch of a $\pm$.
- A function that is not one-to-one needs a restriction first; without one it has no inverse function.

Example: $f(x) = \sqrt{x-1} + 2$. Find f^{-1} in completed-square form.

Step 1. Swap the variables and undo the operations from the outside in, keeping track of where the square root was non-negative.

Step 2. Solving gives $x^2 - 4x + 5$, which is $(x-2)^2 + 1$, valid on $x \geq 2$ because f never outputs less than 2.

Try it: $f(x) = \sqrt{x+3} - 1$. Find f^{-1} in completed-square form.

check: $f^{-1}(f(x)) = x$